



CS 178 Visual Analytics: Final Project

Initial Report Due: Monday 4/13/2026 11:59pm

Final Project Due: Sunday 5/4/2026 11:59 PM

Live Demo! (In Class) Monday 5/5/2026 3:30-5:30pm (Final Exam Period)

Description: The goal of your final project is to apply your knowledge about visual analytics to help the MBTA analyze their data and find out surprising and interesting trends and patterns!

The Challenge: You will be working with the 2025 MBTA Rapid Transit [Headways](#):

This file contains the headways, or time between the departure time of the current trip and the departure time of the previous trip from the same stop, for rapid transit vehicles. It includes all heavy rail and light rail rapid transit lines. For Red and Green lines, both trunk (e.g. all Green line trains) and branch (e.g. Green-B trains) headways are included. For lines without branches (e.g. Blue line), branch route ID and branch headway will be NULL. Due to track circuit or other data issues, data is not guaranteed to be complete for any stop or date.

The data and the description of the data are available [here](#). Your task is to “What, if anything, makes the Green Line Extension to Tufts different from the other lines.” Note that you’re welcome to use additional datasets found online (e.g., using the 2024 headways dataset to compare between the two years, or weather data to correlate weather pattern with transit schedules). Whether or not you do so should depend on the hypotheses that you pose.

Assignment

Part 1: Initial Report (due: Monday 4/13/2026 11:59pm)

- Familiarize yourselves with the dataset
- Generate a report that includes:
 - **Project Title** and team members, including names and email.

- **Abstract** (1-2 paragraphs): The abstract should include your motivation for this project, what you are trying to do, and what you want to achieve.
- **Data Description** (1-2 paragraphs): Summarize the data. Show that you have performed an initial exploration of the data.
- Definition of **Goals and Tasks** of your final project (1-2 pages). Show that, in addition to having explored the data, you understand the goals and tasks of your challenge. Pose (at least) one hypothesis of what your analysis is about, then describe how you plan to prove or disprove it.
- A **Feature List**. Describe tasks, visualizations, tools, analyses, etc. that you will showcase in your final project. Label them as must-have, good-to-have, and optional.
- A **Project Timeline** (with milestones when you are planning to finish which feature)
- **Team Roles**: who's going to be doing what in your team?

Part 2: Answer Sheet (Due: **Monday 5/4/2025 11:59 PM**)

- Provide a document answer the following:
 - Tools Used: Provide a list of tools used. Examples:
 - Tableau
 - Excel
 - Spinviz, developed by the University of West Birmingham CS 459 Information Visualization class, taught Spring 2011 by Dr. Smith, and adapted by the student team for the challenge.
 - Question 1: Characterize the data. What aspects of this data could serve as units of analysis? How do those components relate to one another? Please limit your answer to 8 images and 500 words.
 - Provide your answer and corresponding images.
 - Question 2: Using this data set try to identify relationships between headways and the other variables available to you. Make an argument for why the relationships you find might exist. Please limit your answer to 8 images and 500 words.
 - Provide your answer and corresponding images.
 - Question 3: What are potential issues or shortcomings of this dataset? Consider how the data was collected. Are there any places in the data you think there might be errors? Provide evidence for these relationships. Please limit your answer to 8 images and 500 words.
 - Provide your answer and corresponding images.

Part 3: Video Demo (Due: **Monday 5/4/2025 11:59 PM**)

- We suggest that the video is about 5 minutes in length, but feel free to take more time if needed.
- We would suggest that you either:

- Provide narration/voice-over – at least at the key moments so that we understand what we’re seeing, or
 - Provide text captions in the video
- Your video should demonstrate how you use your system to answer your hypotheses and/or the questions in the “Answer Sheet” (the two shouldn’t be too different).
 - Be sure to make both the “process” (of using your system) and the “finding” clear
- [Optional] Feel free to put in all the info about your system/solution. For example, the architecture (e.g., using a database backend), the pre-computation or pre-processing, the cool interactions you designed, etc.
- Note that I use the term “system/solution” to reinforce the idea that the primary contribution or novelty of your final project doesn’t have to be a visualization interface itself but can be a holistic “solution” (with emphasis on data analysis, pre-computation, etc.). See the FAQ below for more info.

*Part 4: Live Demo and Presentation (On: **Tuesday 5/5/2025 3:30-5:30 PM**)*

- You will present and give a live demo of your completed system during the final exam period for this class (on May 5 at 3:30pm, following the H+ block of Tufts’ Final Exam Schedule). The presentation will take place in the same room as class.
- You will have about 8-10 minutes for your presentation (120 minutes / 11 teams, plus setup, transitioning times, etc.). We will kick you off the stage if you exceed your time limit.
- **Presentation Requirements:** Because each team has limited time to present, you need to use that time wisely! Your presentation should contain the following:
 - Title page (1 page)
 - Project (or team) name
 - Names of members
 - Abstract or Hypothesis page (1 page)
 - Motivation, goal, or the hypotheses of your project
 - Highlight the key question/hypothesis that you’ll use in your demo later
 - Note: Do not use this time to talk about the data or the problem itself. We know that information already.
 - Content page(s)
 - Present how you solved this challenge (what pre-computation you did, what is the architecture of your system, etc.).
 - Let us know if and where you used AI to help with the development
 - Do not to be exhaustive. Focus on the key elements of your project that made your solution successful!
 - Live Demo!!

- We suggest a demo where you use your system to solve one of the hypotheses (you probably won't have time to do more than that).
- However, this is just a suggestion. You should demo your system in whatever way that best showcases your solution.
- Finding page (1 page)
 - Conclude with whether your finding supports or refutes your hypotheses.
 - Not all hypotheses will be supported. Finding negative results can be just as interesting.
- (Optional) Discussion
 - This is somewhat dependent on your finding / solution, but some possible discussion points include:
 - How confident are you in your finding? Why?
 - What aspects of your solution are critical to your success?
 - Are there things that you wish you had done differently?
 - Do you have insights into this challenge that you can share?
- Conclusion (1 page)
 - What's the most important takeaway from your project?

Submission:

- By **Monday 4/13/2025 11:59pm**, submit:
 - A PDF of your Initial Report
- By **Monday 5/4/2025 11:59 PM**, submit the following 4 items:
 - A PDF of your completed "Answer Sheet"
 - A video where you demo your system/solution.
 - A complete zip file of your source code, material, etc.
 - The purpose of this isn't so that we can re-run your system – we understand that this is not likely going to happen if you use a database or run your own webserver, etc. Instead, think of this as an archive – the same files you would have on, say, a Github repo.
- By **Tuesday 5/5/2025 3:29 PM (before class)**, submit:
 - Your slides

FAQ:

- I found some examples of similar analyses! Can I draw inspiration from them?
 - Please do not seek out the answer to the VAST Challenge. They are available online, but we trust that you will refrain from doing so. This project really isn't any fun if you know the answer before you start.
- How will my final project be graded?
 - Your final project will be graded based on: (1) your visual analytics system, (2) your ability to use your system to solve the challenge, and (3) your demo and presentation.

- To be more specific on point 1, your system should have both a computational (automation) component and an interactive interface. Further, it should have the ability to help the user discover the ground-truth answers.
- What should my visual analytics system look like?
 - By now, you have seen academic prototypes and commercial visual analytics systems. You should use those as design inspirations.
 - Keep in mind that the purpose of this system is for an analyst to investigate the challenge (and not for a wide, public audience who lack technical expertise). In this sense, first and foremost you should be developing a tool that can help you solve the problem! If your tool also has a communication component (i.e., if your grandparents can use it), that's a bonus.
- How should I design and develop my visual analytics system?
 - Keep in mind that this final project involves both exploration and confirmation. In this sense, your visual analytics system should have the capability to help the user perform **data exploration and hypothesis confirmation**. Exactly what the system looks like will be up to you!
- How much data processing and precomputation should I do?
 - We highly encourage you to use any tools and techniques you have learned throughout this course. This means that you are welcome to throw your data into Tableau, Pandas, scikit-learn, R, DuckDB, etc. to do whatever exploration, analyses, pre-processing, data-cleaning, etc. you need to do.
- What if my solution is more “computational” than “visual”?
 - Visual analytics systems are a hybrid of automation and (human) interaction. Under this umbrella, your system can be more of one and less of the other. In other words, if your solution has a strong algorithmic or computational contribution (e.g. using ML techniques to pre-process data and find clusters or outliers that you incorporate into the original data) but less novelty in the visualization, that's perfectly fine as a final project. Just make sure that there's still a visualization component that helps the user analyze the outcomes of the algorithm.
- How robust or generalizable does my system need to be?
 - For your final project, we do not require that your tool needs to apply to other data or problems beyond the intended use. In terms of robustness, we also don't care if your system crashes on unexpected inputs. Chances are that we will not ever use your tool (we'll just watch you use your tool during the demo), but you should still be prepared if we ask you to perform certain actions during your demo.
- What happens if we deviate from our original ideas (from the Initial Reports)?
 - It's perfectly normal if, during the course of designing and developing your visual analytics system/solution, you find out that your initial ideas

or approach to the challenge don't work. You are free to deviate or modify your original ideas.

- What about the use of AI/LLM?
 - First, you are free to use whatever AI assistance you need to complete your final project.
 - Considering that this is a CS course, we are still requiring you to “build” a system (that's not just a wrapper around ChatGPT). If your system includes an LLM, we're ok with that. Admittedly, we have not dealt with this before, so we're also somewhat curious as to what new types of AI-assisted visual analytics systems you might come up with (and whether these systems can solve these challenges more efficiently and effectively).
 - Finally, unfortunately, we do not have the budget to pay for your tokens. If you want to use these AI systems, either use off-the-shelf pre-trained models, get a student license, or pay for the API access yourself.